

4.8.6

Response to a causal exponential sequence -

$$X[n] = e^{j\omega n} u[n].$$

$$Y[n] = \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} u[n-k]$$

$$= \left( \sum_{k=0}^n h[k] e^{-j\omega k} \right) e^{j\omega n} \quad \text{as } u[n-k] = 0 \text{ for } k > n$$

Rewrite the eq<sup>n</sup>.

$$Y[n] = \left( \sum_{k=0}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n} - \left( \sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n}$$

$$= H(e^{j\omega}) e^{j\omega n} - \left( \sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n} \quad n \geq 0$$

$$Y_{ss}[n] = H(e^{j\omega}) e^{j\omega n} \rightarrow \text{steady-state response.}$$

$$Y_{tr}[n] = - \left( \sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n} \rightarrow \text{transient response}$$

4.68 Consider an LTI DTs with an impulse response  $h[n] = (0.4)^n u[n]$ . Determine the freq<sup>y</sup> response  $H(e^{j\omega})$  of the system and evaluate its value at  $\omega = \pm \pi/4$ .

- What is the steady-state response of the system for an input  $X[n] = \sin(\pi n/4) u[n]$ ?

Sol<sup>n</sup>:  $h[n] = (0.4)^n u[n]$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k} = \sum_{k=0}^{\infty} (0.4)^k e^{-j\omega k}$$

$$= \sum_{k=0}^{\infty} (0.4)^k e^{-j\omega k} = \sum_{k=0}^{\infty} (0.4 e^{-j\omega})^k = \frac{1}{1 - 0.4 e^{-j\omega}}$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.4(\cos\omega - j\sin\omega)} = \frac{1 + 0.4(\cos\omega + j\sin\omega)}{1 - 0.4^2(\cos\omega - j\sin\omega)^2}$$

$$= \frac{1 + 0.4(\cos\omega + j\sin\omega)}{1 - 0.16(\cos^2\omega + \sin^2\omega - 2j)}$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.4e^{-j\omega}} = \frac{1 + 0.4e^{-j\omega}}{(1 - 0.4e^{-j\omega})(1 + 0.4e^{-j\omega})}$$

$$= \frac{1 + 0.4e^{-j\omega}}{1 - 0.4^2 e^{-j2\omega}}$$

$$= \frac{1 + 0.4e^{-j\omega}}{1 - 0.16}$$

$$= \frac{(1 + 0.4\cos\omega) - j0.4\sin\omega}{((1 - 0.4\cos\omega) + j0.4\sin\omega)((1 - 0.4\cos\omega) - j0.4\sin\omega)}$$

$$= \frac{(1 - 0.4\cos\omega) - j0.4\sin\omega}{(1 - 0.4\cos\omega)^2 - j^2 0.4^2 \sin^2\omega}$$

$$= \frac{(1 - 0.4\cos\omega) - j0.4\sin\omega}{1 - 2(0.4\cos\omega) + 0.16\cos^2\omega + 0.16\sin^2\omega}$$

$$= \frac{(1 - 0.4\cos\omega) - j0.4\sin\omega}{1 + 0.16(\cos^2\omega + \sin^2\omega) - 0.8\cos\omega}$$

$$= \frac{(1 - 0.4\cos\omega) - j0.4\sin\omega}{1.16 - 0.8\cos\omega}$$

$$H(e^{j\omega}) = \frac{1 - 0.4\cos\omega}{1.16 - 0.8\cos\omega} - j \frac{0.4\sin\omega}{1.16 - 0.8\cos\omega}$$

Magnitude Response:  $|H(e^{j\omega})| = \sqrt{A^2 + B^2} = \sqrt{H(e^{j\omega})H(e^{-j\omega})}$

$$|H(e^{j\omega})| = \sqrt{(1 - 0.4e^{-j\omega})(1 - 0.4e^{j\omega})}$$

$$= \sqrt{1 - 2(0.4)e^{j\omega} + (0.4)^2}$$

$$= \sqrt{1 - 0.4e^{j\omega} - 0.4e^{-j\omega} + 0.16}$$

$$\checkmark |H(e^{j\omega})| = \sqrt{1.16 - 0.8 \cos \omega}$$

$\checkmark$  Phase response,  $\theta(\omega) = \tan^{-1}\left(\frac{B}{A}\right) = \tan^{-1}\left(\frac{H_{im}(e^{j\omega})}{H_{re}(e^{j\omega})}\right)$

$$= \tan^{-1}\left(\frac{-0.4 \sin(\omega)}{1.16 - 0.8 \cos \omega}\right)$$

$$= \tan^{-1}\left(\frac{0.4 \sin \omega}{1 - 0.4 \cos \omega}\right)$$

Now,  $\omega = \pm \pi/4$

$$H(e^{\pm j\pi/4}) = |H(e^{\pm j\pi/4})| e^{j\theta(\pm \pi/4)}$$

$$= 1.2977 e^{\pm j0.3735}$$

Now, for an input  $x[n] = \sin(\omega_0 n)$ , the steady-state output is given by -

$$\checkmark y[n] = |H(e^{j\omega_0})| \sin(\omega_0 n + \theta(\omega_0))$$

$$\omega_0 = \pi/4, \quad y[n] = |H(e^{j\pi/4})| \sin\left(\frac{\pi}{4}n + \theta\left(\frac{\pi}{4}\right)\right)$$

$$= 1.2977 \sin\left(\frac{\pi}{4}n + 0.3735\right) \checkmark$$

#### 4.8.7: The concept of filtering (LTI DTS)

Let

$$|H(e^{j\omega})| \cong \begin{cases} 1 & 0 \leq \omega \leq \omega_c \\ 0 & \omega_c \leq |\omega| < \pi \end{cases} \quad (1)$$

Input:  $x[n] = A \cos \omega_1 n + B \cos \omega_2 n$   $0 < \omega_1 < \omega_c < \omega_2 < \pi$ .

Because of linearity-

Output:  $y[n] = A |H(e^{j\omega_1})| \cos(\omega_1 n + \theta(\omega_1)) + B |H(e^{j\omega_2})| \cos(\omega_2 n + \theta(\omega_2))$

using eq<sup>n</sup> (1).

$$y[n] = A |H(e^{j\omega_1})| \cos(\omega_1 n + \theta(\omega_1)) \rightarrow \text{LPF}$$

Ex<sup>o</sup> 4.33: The input signal consists of a sum of two cosine sequences of angular frequencies 0.1 rad/samples and 0.4 rad/samples.

Design a HPF.

Sol<sup>n</sup>: For simplicity, we assume the filter to be an FIR filter of length 3 with an impulse response

$$h[0] = h[2] = \alpha_0, \quad h[1] = \alpha_1$$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k] = h[0] x[n] + h[1] x[n-1] + h[2] x[n-2]$$

$$\Rightarrow y[n] = \alpha_0 x[n] + \alpha_1 x[n-1] + \alpha_0 x[n-2]$$

Our goal is to choose suitable values for the filter parameters,  $\alpha_0$  and  $\alpha_1$ , so that the output of the filter is a cosine sequence with a frequency of 0.4 rad/samples.

$$\begin{aligned} H(e^{j\omega}) &= h[0] + h[1] e^{-j\omega} + h[2] e^{-j2\omega} \\ &= \alpha_0 + \alpha_1 e^{-j\omega} + \alpha_0 e^{-j2\omega} \\ &= \alpha_0 (1 + e^{-j2\omega}) + \alpha_1 e^{-j\omega} \end{aligned}$$

$$H(e^{j\omega}) = 2\alpha_0 \left( \frac{e^{j\omega} + e^{-j\omega}}{2} \right) e^{-j\omega} + \alpha_1 e^{-j\omega}$$

$$= (2\alpha_0 \cos \omega + \alpha_1) e^{-j\omega}$$

$$|H(e^{j\omega})| = |2\alpha_0 \cos \omega + \alpha_1|$$

$$\theta(\omega) = -\omega + \beta, \quad \text{where } \beta = 0, \text{ when } 2\alpha_0 \cos \omega + \alpha_1 > 0$$

$$\beta = \pi, \quad \text{if } 2\alpha_0 \cos \omega + \alpha_1 < 0$$

In order to stop low frequency  $\omega_1 = 0.1$  the  $|H(e^{j\omega})| = 0$   
 In order to pass high  $\omega_2 = 0.4$  the  $|H(e^{j\omega})| = 1$

$$\left. \begin{array}{l} 2\alpha_0 \cos(0.1) + \alpha_1 = 0 \\ 2\alpha_0 \cos(0.4) + \alpha_1 = 1 \end{array} \right\} \begin{array}{l} \alpha_1 = \cancel{-6.76195} - 6.76195 \\ \alpha_0 = 13.456335 \end{array}$$

$$// \quad y[n] = -6.76195 (x[n] + x[n-2]) + 13.456335 x[n-1]$$

$$\text{with } x[n] = \frac{1}{2} (\cos(0.1n) + \cos(0.4n)) \mu[n]$$